

SHAHAJADI SADIA AFSANA

Data Analyst · Data Engineer · ML Engineer

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SUMMARY

Data professional with 3+ years across analytics and machine learning at a fast-growing agri-fintech company. I design and automate ETL pipelines, analyze large operational, financial, and geospatial datasets in Python and SQL, and ship decision-ready reporting for cross-functional teams. Hands-on experience across the ML pipeline EDA, feature engineering, modeling, and evaluation, including RAG and LLM-based applications.

EXPERIENCE

iFarmer Asia Ltd.

Dhaka, Bangladesh

Data Analyst

Mar 2024 – Present

- Built the iFarmer interactive mapping system - [Krishi Map](#) end-to-end from data collection to Folium visualization, now used for business decision-making across every team; most impactful work to date.
- Built the farmer-profiling logic and data-prep pipeline for a credit-scoring system, supporting lending decisions that achieved a ~98% repayment rate.
- Built a weather early-SMS-alert system delivering rainfall alerts to farmers ahead of time, integrated into the iFarmer platform.
- Building a flood early-warning system on the Google Flood Hub API that alerts farmers before flooding occurs.
- Integrated Google Places, Geocoding, and Flood Hub APIs into the iFarmer map, unifying government datasets with internal farmer data into a single geospatial view.
- Analyzed maize and paddy price patterns using time-series techniques, surfacing trends that informed company pricing and procurement decisions.
- Built the geospatial data-processing pipeline and contributed to the ML model for a satellite-based farm-monitoring system.
- Contributed to the knowledge base powering **Krishi Master**, an LLM-based disease-detection tool for farmer-facing diagnosis.
- Stored, managed, and analyzed datasets that powered a fertilizer-recommendation tool, helping farmers cut input costs and raise production.

Associate Data Scientist

Dec 2023 – Mar 2024

- Built the iFarmer interactive map (individual) and ran ad-hoc analyses that fed business decisions.
- Built the farmer-profiling logic for a credit-scoring system, helping assess loan eligibility (~90% repayment rate).

Data Analyst (Intern)

Sep 2023 – Dec 2023

- Worked alongside senior team members on data tasks, building hands-on data-analysis skills.

Catalyst Solutions (Adiva Graphics)

Dhaka, Bangladesh

Market Research Data Analyst

Feb 2023 – Jul 2023

- Analyzed and visualized clients' business data, turning it into insights that guided their business decisions.

EDUCATION

M.Sc. in Computer Science and Engineering (Data Science)

2026 – Present

BRAC University

B.Sc. in Computer Science and Engineering

2018 – 2023

Ahsanullah University of Science and Technology (AUST)

Higher Secondary Certificate

2015 – 2017

Holy Cross College

Secondary School Certificate

2013 – 2015

Bangladesh Bank High School

RESEARCH

Machine Learning-based Investment Decision Support System: Candlestick Chart Pattern Identification and Price Breakout Prediction

Using Dhaka Stock Exchange (DSE) data from 2000 to 2020 and five common machine learning models, this research detects harmonic patterns (Gartley, Butterfly, BAT, Crab). It identifies the target value after a harmonic pattern forms. The dataset was trained using an Artificial Neural Network and a Multilayer Perceptron.

Advances in Mental Health Detection Using Machine Learning and Deep Learning with Behavioral Data: A Comprehensive Synthesis

Research Poster, CARSIT Research Masterclass 2026 (Supervisor: Karim Mohammed Rezaul)

Benchmarking machine learning and deep learning methods for early mental-health detection using survey, social media, and health-record data; analyzed predictive biomarkers, implementation challenges, and ethical considerations.

PERSONAL PROJECTS

FIFA 2026 — AI-Powered Analytics | *Python, XGBoost, scikit-learn, Monte-Carlo, Groq, Streamlit*

A fun World Cup analytics project featuring match prediction, Elo ratings, Monte Carlo tournament simulations, and an AI-powered chatbot, deployed on Hugging Face. · [Repo](#) · [Live](#)

ML End-to-End Workflow — Price Prediction | *Python, scikit-learn* · [Repo](#)

Full ML pipeline: EDA, leakage-free preprocessing, feature engineering, cross-validated model selection, and evaluation.

AI Document System — End-to-End RAG Pipeline | *Python, FAISS, sentence-transformers, Groq* · [Repo](#)

Full RAG pipeline ingesting messy PDFs/scans, chunking, embeddings, FAISS retrieval, and grounded draft generation with citation enforcement and an operator-feedback loop.

RAG Apps — PaperMind | *Python, Groq API, Streamlit* [PaperMind](#)

Retrieval-augmented generation apps: a document-querying RAG tool deployed as a Streamlit interface.

LLM Apps with Gemini & Groq | *Python, Groq-Gemini API, Streamlit* · [Note-Generating Agent](#) | [CodeLens AI](#) | [LearnMate AI](#) | [Gemini Chatbot](#) | [Improve Proficiency](#) | [Prompt to Response](#) | [PrepMate-AI](#)

AI suite, chatbots, and a note-generating agent built on Gemini and Groq APIs.

ML Projects | *Python, scikit-learn* · [Repo](#)

Classification (Heart Disease, Titanic) and regression (Big Mart, Car, House, Insurance, Calories, Gold) models with feature engineering and evaluation.

Time-Series Forecasting — Gold Price | *Python, statsmodels* · [Code](#)

Gold price forecasting with trend/seasonality analysis and forecast evaluation.

ACADEMIC PROJECTS

Face Mask Detection | *Python, PyTorch* · [Repo](#)

A binary classification model detecting whether a person is wearing a mask.

Heart Attack Possibility Prediction | *Python, NumPy, Pandas* · [Repo](#)

ML project predicting heart-attack risk using multiple models.

Music Genre Classification | *Python* · [Repo](#)

Model that classifies music into genres from mixed audio (GTZAN dataset).

SKILLS

Programming Languages: Python, JavaScript, SQL, C, C++, Java

Data Analysis: EDA, feature engineering, data cleaning, statistical analysis, trend & pattern analysis, time-series techniques

Data Engineering: Web scraping, ETL pipelines, automated data collection, API integration (FastAPI)

Machine Learning, RAG pipelines, LLM applications

Libraries: Pandas, NumPy, scikit-learn, PyTorch

Visualization: Folium, Matplotlib, Plotly, Seaborn, Power BI, Tableau, Streamlit, Excel (tables, dashboards)

Geospatial: Folium, GeoParquet, GeoJSON, Google APIs, spatial indexing, reverse geocoding, satellite data

IDE: GitHub, Jupyter, Google Colab, Kaggle, VS Code, PyCharm

Operating System: Windows, Ubuntu

STUDY & CERTIFICATIONS

Phitron AI/ML (in progress, 2026–2027): deep learning (PyTorch, CNNs, RNNs/LSTMs, Transformers), NLP, Generative AI (LangChain, RAG), with a research/thesis track.

Data Analyst Bootcamp — Alex The Analyst (in progress).

Research Masterclass — Centre for Applied Research in Software & IT (CARSIT), 2026 · ID 251181 [Show credential](#)

Teaching Experience Certificate — Dhaka Tuition, 2026 · ID DT/ATC/2026 [Show credential](#)

HONORS & ACTIVITIES

Senior Sub Executive — AUST Cultural Club (AUSTCC), 2023. Organized cultural programs, led team coordination, and event management.

Sports Team Leader - Trojan Team, Holy Cross College.

Mathematics Olympiad, Holy Cross College Science Club, 2015. [Certificate](#)

Divisional Science Olympiad (Junior Group), Bangladesh Academy of Sciences, 2015. [Certificate](#)

Former Guide - Bangladesh Girl Guides Association.

Ranked 1st in class consistently from Grade 1 through Grade 10.

REFERENCES

Dr. Mohammad Shafiul Alam — Professor, Dept. of CSE, AUST, Dhaka · shafiul.cse@aust.edu

Mr. Md. Khairul Hasan — Associate Professor, Dept. of CSE, AUST, Dhaka · khairul271276@aust.edu